

Switching rates of selected DPT cases

Signed average slopes; DUT terminal voltage and current.

17 September 2026

L (nH)	Setpoint (A)	External R_{on}/R_{off} (Ω)	Second turn-on		First turn-off	
			dv/dt (V/ns)	di/dt (A/ns)	dv/dt (V/ns)	di/dt (A/ns)
1	15	0.5/0.5	-12.48	14.62	5.38	-0.91
1	30	0.5/0.5	-10.95	21.08	10.61	-3.25
1	45	0.5/0.5	-9.47	24.91	14.89	-5.57
3	15	0.5/0.5	-12.16	12.22	5.46	-0.98
3	30	0.5/0.5	-10.51	15.06	9.87	-2.45
3	45	0.5/2	-8.70	18.92	20.07	-10.00
5	15	4/1	-6.72	4.83	5.36	-0.85
5	30	4.5/1.5	-5.61	6.67	14.21	-5.13

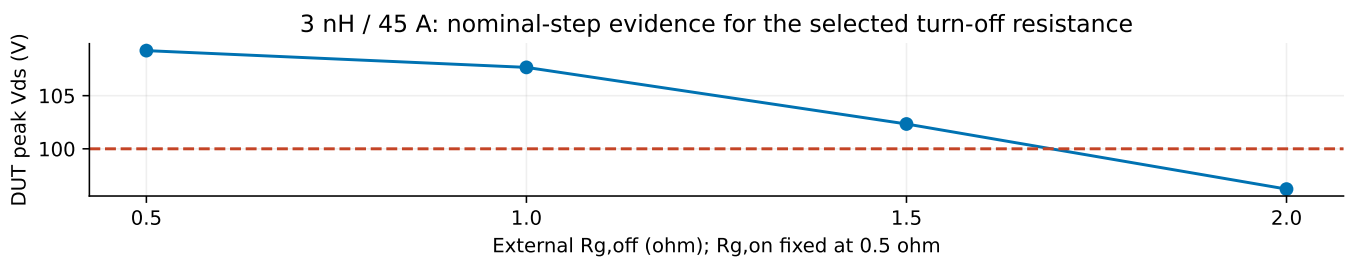
Second turn-off and actual current references

L (nH)	Setpoint (A)	$I_{on2}/I_{off1}/I_{off2}$ (A)	dv/dt (V/ns)	di/dt (A/ns)
1	15	14.76/15.02/18.45	6.10	-1.04
1	30	29.72/30.02/33.37	11.36	-3.61
1	45	44.69/45.02/48.30	17.40	-10.83
3	15	14.77/15.02/18.46	6.79	-1.45
3	30	29.73/30.02/33.38	14.73	-10.56
3	45	44.71/45.02/48.32	19.18	-9.98
5	15	14.72/14.97/18.34	6.79	-5.46
5	30	29.68/29.96/33.26	8.73	-2.53

No selected pair exists for 5 nH / 45 A under the previous 100 V screening criterion; no slope is presented as a passing setting for that case.

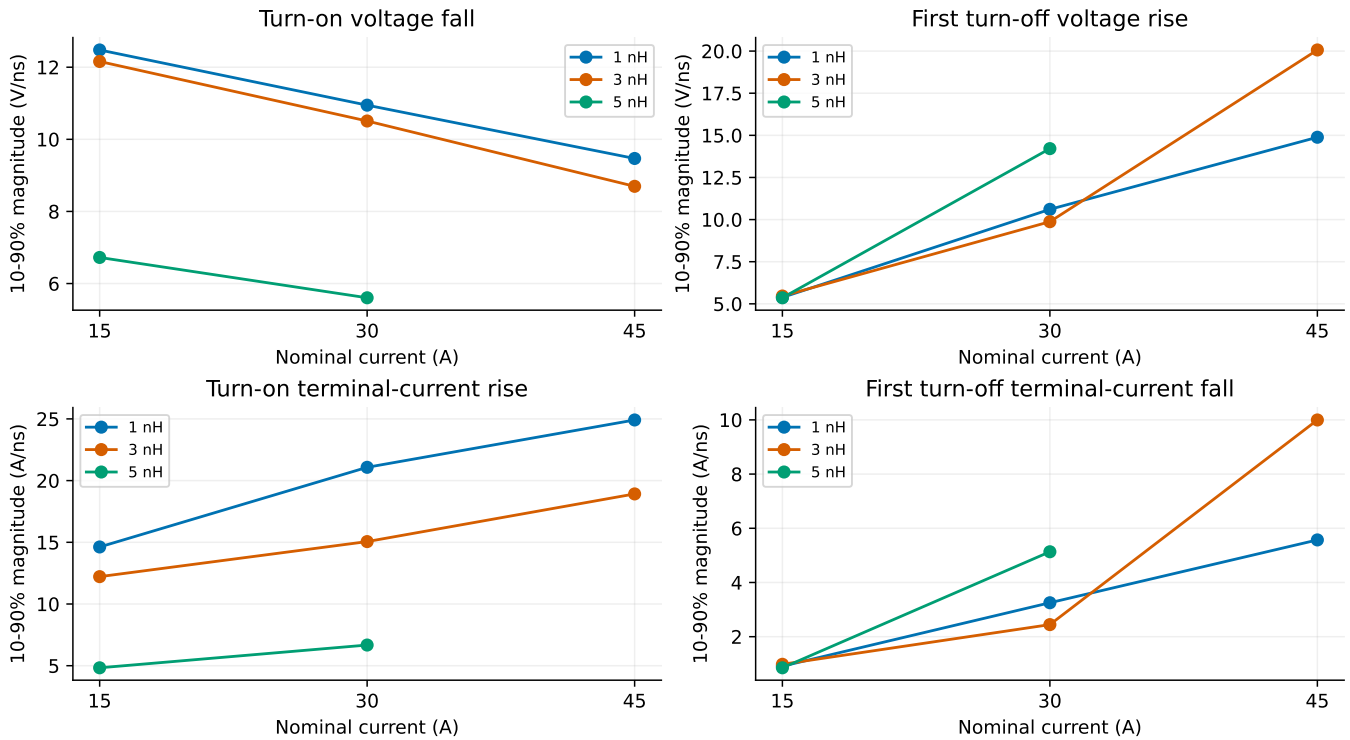
Why can R_{off} exceed R_{on} ? Only 3 nH / 45 A has that external-resistor ordering. Five cases have equal 0.5 Ω external resistors, but the driver adds 0.7/0.4 Ω : the drive paths are approximately 1.2/0.9 Ω , before internal gate resistance. The 5 nH selections also have lower R_{off} .

A strong pull-down helps hold the gate off against Miller current. However, current commutation and ringing in a parasitic inductance can limit turn-off speed. The search minimized $R_{on} + R_{off}$ (ties: lower R_{on}) subject to DUT peak voltage, not energy, switching time, or a required resistance ratio. The plot below shows the actual voltage trade-off; it does not prove a monotonic gate-resistance/slopes relationship.

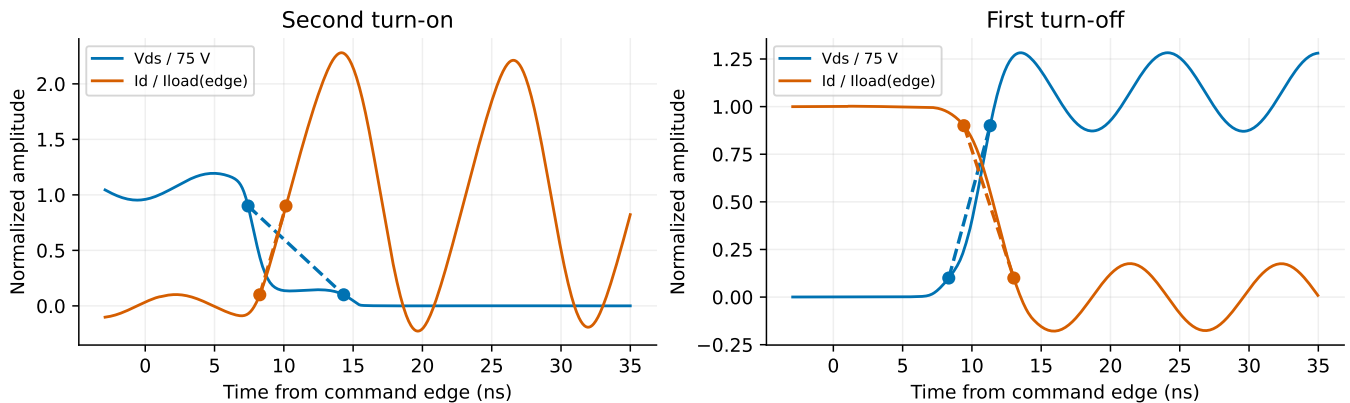


Gate-drive background: EPC AN003, pp.2–3, discusses Miller turn-on, gate-loop damping and their trade-off. EPC: Using Enhancement Mode GaN-on-Silicon Power FETs.

Measured rates and extraction windows



Magnitudes are plotted for readability; the tables retain signs. Each point uses its own selected resistors, so these are not fixed-resistance inductance sweeps.



Example: 3 nH / 45 A, external 0.5/2 Ω . Dashed chords and dots mark the extracted voltage and current intervals. Early incomplete current excursions are excluded.

Definition: voltage thresholds are 7.5 and 67.5 V (10–90% of the nominal 75 V link). Current thresholds are 10–90% of I_{load} at each command edge. Rates are signed $\Delta V / \Delta t$ and $\Delta I / \Delta t$. Within 200 ns after the command, take the first completed directed transition, using the last start-threshold crossing preceding its end-threshold crossing. Linear interpolation determines crossing times.

Interpretation: $I_D = I(V_{sense})$ includes device capacitive/displacement current. It is not internal channel current, nor necessarily the current in L_{layout} . These are transition averages, not peak derivatives or settled-current slopes; ringing remains visible. Rates use the saved 0.125 ns runs and have not received a separate slope-convergence study. Settings remain 75 V, 100 °C and +5/-4 V gate rails; the original voltage-screening limitations still apply.

Data and crossing times: `dut_gate_search/switching_rates.csv`. Reproduce: `python build_switching_rates_report.py`. The GUI and original three-page report are unchanged.